

When Explanations Outlive Their Data: Faithfulness Decoupling in Graph-Attention MARL Fleet Dispatch under Telemetry Degradation

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ABSTRACT

Graph-attention multi-agent reinforcement learning (GAT-MARL) exposes weights that are easy to present as explanations of fleet-dispatch decisions. In an operational fleet, however, those weights may refer to stale vehicle telemetry. This study tests whether attention explanations remain trustworthy under tunnel-triggered signal loss. SUMO retains the true vehicle state, while an observation-layer degradation module freezes the last valid position and speed for 10, 20, 30, or 60 seconds. Clean-trained and degradation-trained GAT policies are each trained with three independent seeds, selected on validation demand, and tested with eight evaluation seeds and three episodes per condition. Explanation quality is measured by decision-level explanation faithfulness (DEF) with type-matched random occlusions, and stale-data exposure is measured by weighted attention mass on stale nodes (WAMSN). WAMSN increases with outage duration for all six trained GAT policies (within-policy Spearman ρ 0.078-0.121; Holm-adjusted $p = 0.0004$). In contrast, the exact paired attention shift is positive for two training seeds and negative for one under both training regimes. No run supports a direct decrease in DEF or a faster decline in faithfulness than dispatch performance. The WAMSN-DEF relationship is also seed-dependent. A supporting validity audit shows that request-node occlusion can delete an action, whereas taxi-node occlusion only hides information; type-matched controls are therefore required. The results show that longer outages reliably increase stale-data exposure, but their effect on attention allocation and explanation faithfulness is not stable across trained policies. Attention weights should not be treated as trustworthy explanations without freshness-aware and model-specific validation.

Index Terms: explainable reinforcement learning, graph attention, multi-agent reinforcement learning, fleet dispatch, telemetry degradation, Age of Information, faithfulness.

I. INTRODUCTION

Fleet dispatch depends on relationships among vehicles, passenger requests, and road conditions. Graph neural networks represent these relationships directly, while graph attention networks learn weights over neighbouring nodes [6], [7]. Those weights are often attractive as built-in explanations because they can be displayed without training a separate explainer.

Attention is not automatically a faithful account of a model's decision. Studies in other domains show that attention may be weakly related to feature importance and that different attention maps can support similar predictions [9]-[12]. Fleet systems add a second problem: the input may become stale while the explanation remains visible.

A taxi entering a tunnel may stop transmitting its current position and speed. The dispatcher then continues to use the last valid reading while its Age of Information (AoI) grows [13]. This paper calls the resulting trust risk **faithfulness decoupling**: the

explanation and the evidence supporting the decision may separate, even if dispatch performance appears stable.

The study asks:

1. Is graph attention more faithful than a fair random explanation on clean telemetry?
2. Does tunnel-triggered degradation move attention toward stale vehicle nodes?
3. Does explanation faithfulness decline faster than dispatch behaviour as outage duration increases?
4. Does degradation-aware training make the explanation response more reliable?

The main contribution is a cross-training-seed answer. Thousands of decisions can make a result precise for one checkpoint, but they do not show that the same effect will appear after retraining. The final synthesis therefore treats each independently trained policy as a replicate.

II. RELATED WORK

A. MARL and Graph Attention for Dispatch

Multi-agent reinforcement learning has been used for large-scale fleet management and ride-sharing because vehicles make coupled decisions under shared demand [1], [2]. MAPPO provides a stable policy-gradient framework for cooperative multi-agent tasks [5]. Graph models add an explicit representation of vehicle-request relationships [6], and GAT assigns learned weights to those relationships [7]. Recent dispatch work continues to use graph attention to improve matching and coordination [14], [15]. These studies primarily evaluate dispatch quality, not whether attention remains a trustworthy explanation when telemetry is stale.

B. Attention Faithfulness

The attention-as-explanation debate distinguishes a readable weight map from a faithful account of the evidence that affects the output. Jain and Wallace [9] reported weak agreement between attention and alternative importance measures. Wiegreffe and Pinter [10] argued that the answer depends on the definition and test. Serrano and Smith [11] found that attention is only partly informative, while Liu et al. [12] proposed explicit faithfulness-violation tests.

Perturbation tests commonly remove highly ranked features and compare the output change with random removals [8]. This study adopts that idea but audits the random control because graph nodes have different semantic roles.

C. Telemetry Freshness

AoI measures how long it has been since the newest valid update was generated [13]. Most AoI work asks how stale information affects communication or control. Here AoI is used to describe observation freshness and to weight stale-node attention. The study does not assume that AoI itself causes lower faithfulness.

III. EXPERIMENTAL DESIGN

A. Environment and Data Flow

SUMO [16] simulates a 20-taxi fleet, 50 passenger requests, and 1,200 seconds of traffic in a tunnel-rich road network from Yubei, Chongqing. Demand is divided into training, validation, and held-out test splits.

Tunnel entry is the physical trigger, but degradation occurs at another layer. SUMO continues to update ground-truth movement. The observation layer freezes the last valid position and speed supplied to the policy for the declared outage duration. Each degraded decision is paired with a clean twin generated at the same simulation step.

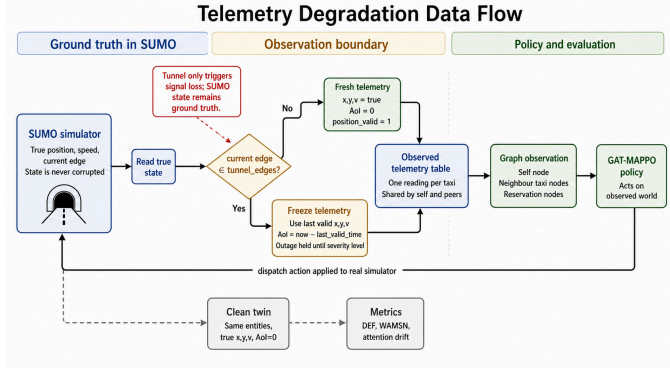


Fig. 1. Tunnel entry triggers the outage; stale telemetry is created only at the observation boundary.

Model	Architecture	Training data	Purpose
B1	MLP-MAPPO	clean	performance context without attention
B2	GAT-MAPPO	clean	primary attention audit
B3	GAT-MAPPO, no AoI input	clean	structural freshness-input control
H5	GAT-MAPPO	30-second tunnel outages	degradation-aware training

Each model is trained for 150 epochs with seeds 42, 43, and 44. Checkpoints are saved every ten epochs and selected by stochastic mean pickups on validation demand; test demand is not used for selection. B2 and B3 are identical under clean evaluation because AoI is zero in both conditions. This does not test an AoI effect under degradation.

D. Degradation Sweep

B2 and H5 are evaluated under clean telemetry and fixed observation-layer outages of 10, 20, 30, and 60 seconds. Each sweep contains:

5 conditions x 8 evaluation seeds x 3 test episodes

The configuration uses freeze corruption and stochastic policy evaluation. All six sweeps pass preflight checks for expected cells, provenance, type-matched controls, chosen-action protection, and increasing empirical degradation.

E. Explanation Measures

DEF compares the output change caused by the explanation's top-k nodes with the change caused by five size- and type-matched random subsets for $k \in \{1, 2, 3\}$. Positive DEF means that the explanation identifies more decision-relevant evidence than its random control. A probability form and an action-protected logit-margin form are recorded.

WAMS measures attention assigned to stale vehicle information:

B. Observation Graph and Policy

Each acting taxi observes itself, up to five nearby taxis, and up to five passenger requests. Taxi nodes contain position, speed, availability, distance, and AoI features. Request nodes contain pickup/drop-off vectors and waiting time. The action space contains no-op plus one action for each request node.

The GAT-MAPPO actor uses node-type projections, two graph-attention layers, and four heads. The actor scores the available actions and the critic supports centralised training with decentralised execution. Attention tensors are returned directly for evaluation.

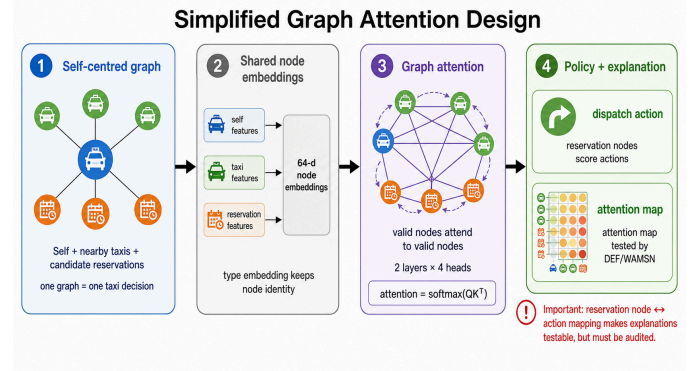


Fig. 2. The attention channel is part of the policy computation and is then audited as a possible explanation.

C. Model Conditions and Training

$$\text{WAMS} = \frac{\sum(\alpha_i * \text{normalised_AoI}_i)}{\sum(\alpha_i)}$$

The paired stale-attention shift compares stale-node attention mass in the degraded observation with its exact clean twin. Unlike WAMS, this paired measure can be positive or negative.

F. Construct-Validity Control

A passenger-request node is also an action candidate. Deleting it can remove the chosen action. A taxi node, by contrast, only supplies contextual information. Uniform random occlusion can therefore alter the action space more often than attention top-k occlusion and create a misleading DEF value.

The final protocol matches each random subset to the same taxi/request composition as the explanation subset. It also records an action-protected variant and counterfactual clamps. The audit supports the primary experiment by ensuring that DEF does not confuse action deletion with hidden evidence.

G. Statistical Analysis

H1 tests whether DEF decreases with outage duration. H2 compares faithfulness and pickup degradation rates. H3 tests whether WAMS increases with duration. H4 tests whether WAMS and DEF are negatively associated within an episode. H5 asks whether degradation-aware training changes these relationships.

Episode-block permutation is used for H1 and H3. H2 uses paired cell-level sign flips. H4 computes Spearman correlation within

each episode before a sign-flip test. Holm correction covers H1-H4 separately for each trained policy. Across training seeds, the paper reports consistency counts and effect ranges, not a pooled population p-value.

Model	Mean pickups	Training-seed range
B1 MLP	13.25	11.50-14.67
B2 GAT	13.29	11.46-14.67
B3 GAT without AoI	13.29	11.46-14.67
H5 degradation-trained GAT	9.56	4.54-12.88

H5 seed 44 is substantially weaker than the other H5 policies. It is retained because removing it would hide a real training outcome. Performance is context for the explanation analysis, not the primary success criterion.

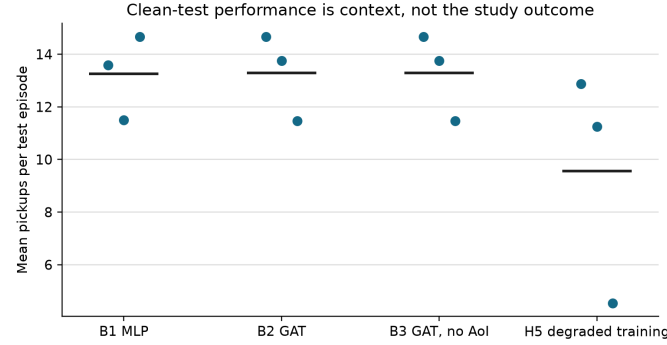


Fig. 3. Clean-test pickups for each independently trained policy.

B. Stale Exposure Increases Consistently

Conditional WAMSN rises from approximately 0.023-0.031 at 10 seconds to 0.071-0.090 at 60 seconds. H3 is supported in every

Model	Seed 42	Seed 43	Seed 44	Positive seeds
B2 GAT	+0.002706	+0.000618	-0.001278	2/3
H5 degradation training	+0.002374	+0.000783	-0.001231	2/3

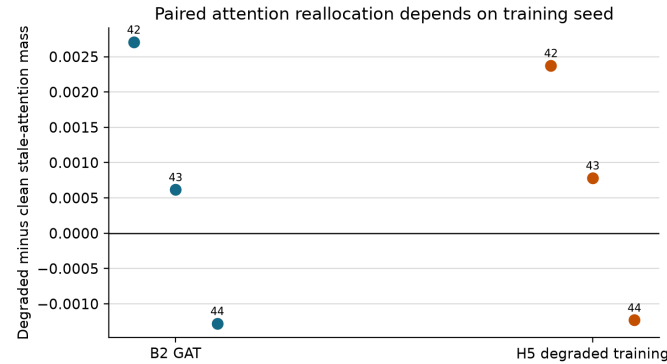


Fig. 5. Degraded-minus-clean stale-attention mass. Seed 44 reverses the direction under both training regimes.

The estimates are precise within each policy because observations are paired, but they disagree across independently trained policies. A universal claim that degradation shifts attention toward stale nodes is therefore not supported.

D. Faithfulness Hypotheses

H1 is unsupported in all six GAT policies: DEF does not decrease with outage duration. H2 is also unsupported throughout:

IV. RESULTS

A. Performance Context

B2 and H5 training seed. Within-policy rho ranges from 0.078 to 0.121, with Holm-adjusted $p = 0.0004$ in all six cases.

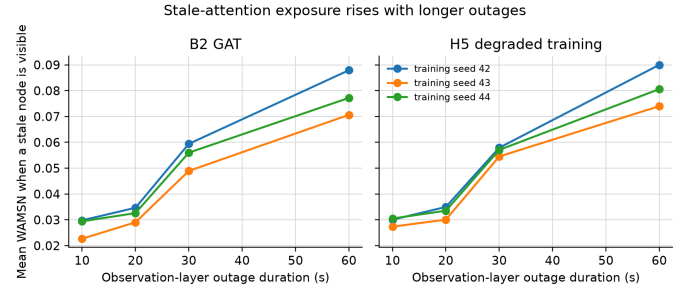


Fig. 4. WAMSN increases with observation-layer outage duration for every trained GAT policy.

This is a manipulation and exposure result. Because WAMSN includes normalised AoI, it must not be interpreted as evidence that AoI reduces faithfulness.

C. Paired Attention Shift Depends on Training Seed

faithfulness does not decline faster than pickups. Clean type-matched DEF remains close to zero, so percentage rate calculations around that denominator are unstable.

H4 is unsupported for every B2 policy. The B2 within-episode WAMSN-DEF correlations are positive (0.036-0.069), contrary to the predicted negative direction. H5 seeds 42 and 43 support a negative association (rho -0.072 and -0.061; adjusted $p = 0.0004$), but H5 seed 44 does not (rho -0.001; adjusted $p = 0.185$). The result is mixed rather than general.

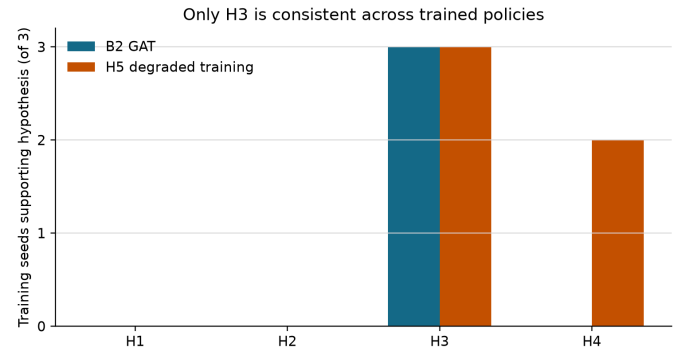


Fig. 6. Number of training seeds supporting each hypothesis after within-policy Holm correction.

Hypothesis	B2	H5	Verdict
H1: DEF decreases with duration	0/3	0/3	not supported
H2: faithfulness declines faster	0/3	0/3	not supported
H3: WAMSN increases with duration	3/3	3/3	consistently supported
H4: greater WAMSN means lower DEF	0/3	2/3	mixed
H5: degraded training gives consistent mitigation	n/a	0/3	not supported

V. DISCUSSION

The experiment supports a narrower interpretation of explanations outliving their data. During an outage, the attention map remains available while some of its vehicle nodes represent earlier states. Longer outages reliably increase stale-data exposure. Yet the exact attention response can reverse after retraining, and no consistent decline in DEF is observed.

This distinction is important. WAMSN is useful for monitoring freshness exposure, but it is not a causal measure of faithfulness. DEF tests decision-relevant evidence, while paired attention shift tests reallocation against a clean twin. Combining them into one claim would overstate the result.

Training-seed variation is an operational risk. A dashboard validated on one checkpoint may behave differently after the same architecture is retrained. Faithfulness and freshness tests should therefore be release checks for each model, not one-time properties assigned to the architecture.

The construct-validity audit is equally practical. In candidate-action graphs, feature removal can also remove an action. Type matching and chosen-action protection are necessary before perturbation scores can be interpreted as explanation quality.

VI. LIMITATIONS

The benchmark covers one district, one freeze corruption, 20 taxis, 50 requests, and sparse tunnel exposure. Three training seeds reveal heterogeneity but are too few for a strong population-level estimate. H5 seed 44 is weak, so competence and explanation response are partly confounded. DEF uses one family of perturbation tests, and WAMSN depends on the selected AoI normalisation. Real telemetry can include delay, packet reordering, partial failure, and noisy recovery, none of which is represented here.

VII. CONCLUSION

Observation-layer outage duration consistently increases stale-attention exposure, but it does not consistently reduce explanation faithfulness or shift attention in one direction across trained policies. Degradation-aware training does not remove this

variation. The study therefore does not claim that AoI causes lower faithfulness. It shows that an attention map can remain plausible after its data become stale and that its response is checkpoint-dependent.

The practical conclusion is direct: graph-attention weights should be treated as model internals unless each released policy passes action-aware faithfulness tests and freshness-aware monitoring. Stable dispatch output is not evidence that the explanation remains trustworthy.

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